FZSoNick

GREEN & SAFE SALT BATTERY

BATTERY SIZING

ACCORDING TO IEEE METHODOLOGY

by G.Trotti

Sizing Tool 2.3

CUSTOMER	VEXTROM / EQUINOR - RAIA
PROJECT	Q2510-1288-00
DATE	12/1/2025
SIZED BY	ARO

178

LOAD TYPE	Variable Power [W]

BATTERY TYPE	250UP40

SYSTEM INFORMATION			
V BUS max	[V]	305	OK
V BUS min	[V]	176	

PERIOD (MIN)		LOAD [W]	
1	1	41110	OK
2	29	37430	OK
3	30	32410	OK
4			OK
5			OK
6			OK
7			OK
R			OK

MAX PERIOD 30h

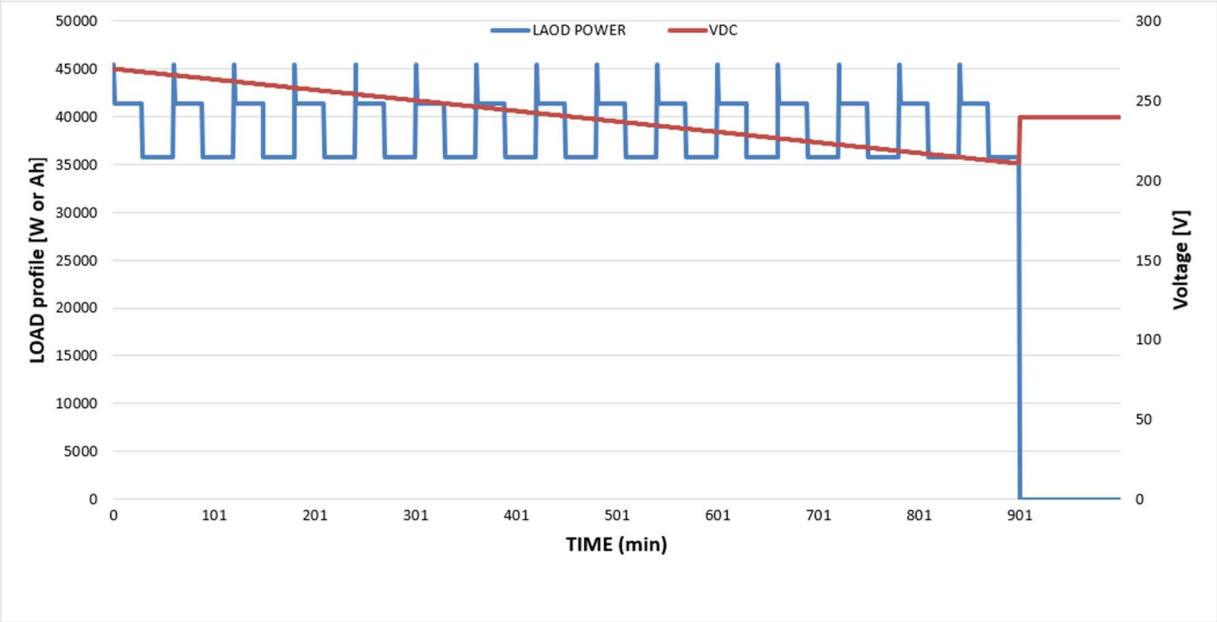
Nominal Energy Output [Wh]	524 720	
DESIGN MARGIN	1.00	OK
Inverter Efficiency [%]	90.51	OK
Battery Energy Required [Wh]	579 737	

IEEE Uncorrected size [Wh]	611 429
IEEE Required Energy [Wh]	611 429

Battery Module Data	
Cell in series	100
Nominal Capacity [Ah]	40
Available Energy [Wh]	9600
Short Circuit Current [A]	250

Total Nominal Available Capacity [Ah]	2520	
Total Nominal Available Energy [Wh]	614 400	
Number of battery modules IEEE	63.7	
Battery modules adjustment	64.0	
Short Circuit Current [A]	15750	
Vmin [V]	210.1	OK
DOD [%]	100	OK
Calculated Recharge Time: Target SOC 95% [h]	8.2	
Calculated Recharge Time - TOC : SOC 100% [h]	11.0	

Total Battery Weight [kg]	7680
Total Battery Volume [dm^3]	5770



BATTERY SIZING

ACCORDING TO IEEE METHODOLOGY

GREEN & SAFE SALT BATTERY

CUSTOMER **VEXTROM /EQUINOR - RAI**
PROJECT **Q2510-1288-00**
DATE **12/02/2025**

PERIOD LOAD	LOAD [W]		CHANGE IN LOAD		DURATION OF PERIOD [MIN]		TIME TO END OF SECTION [MIN]		ENERGY RATING FACTOR	WATTHOURS FOR SECTION	
										POSITIVE	NEGATIVE
Section 1											
1	A1	0	A1-0=	0	M1=	10	T1=M1	0			
									Sub- Total	0,00	0,00
									Total 1	0,00	0,00
Section 2											
1	A1=	0	A1-0=	0	M1=	0	T=M1+M2	0		0,00	0,00
2	A2=	0	A2-A1=	0	M2=	0	T=M2	0		0,00	0,00
									Sub Total=	0,00	0,00
									Total 2	0,00	**0,00**
Section 3											
1	A1=	45420,4	A1-0=	45420,4	M1=	1	T=M1+M2+M3	60	15,82	43113039,44	
2	A2=	41354,5	A2-A1=	-4065,8	M2=	29	T=M2+M3	59	15,82		-3794982,21
3	A3=	35808,2	A3-A2	-5546,3	M3=	30	T=M3	30	15,82		-2632296,98
									Sub Total=	43113039,44	-6427279,20
									Total 3		**611429,34**

Uncorrected Size 611429 Required Energy 611429 DOD MÁX [%] 100

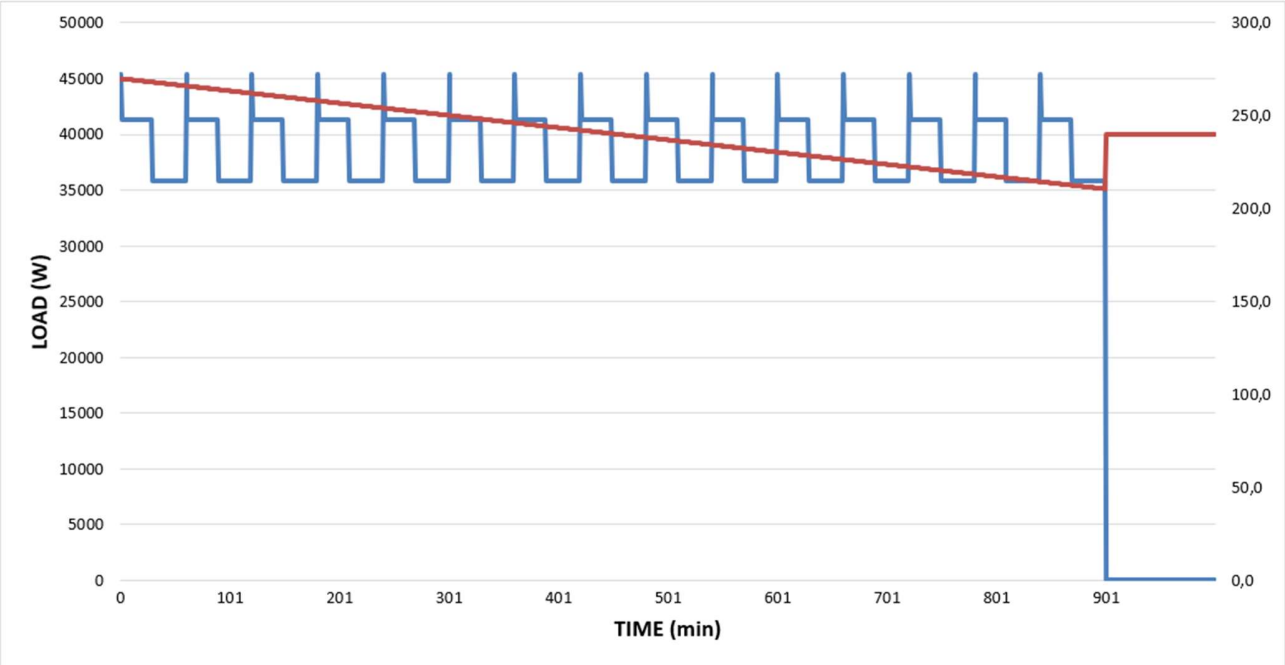
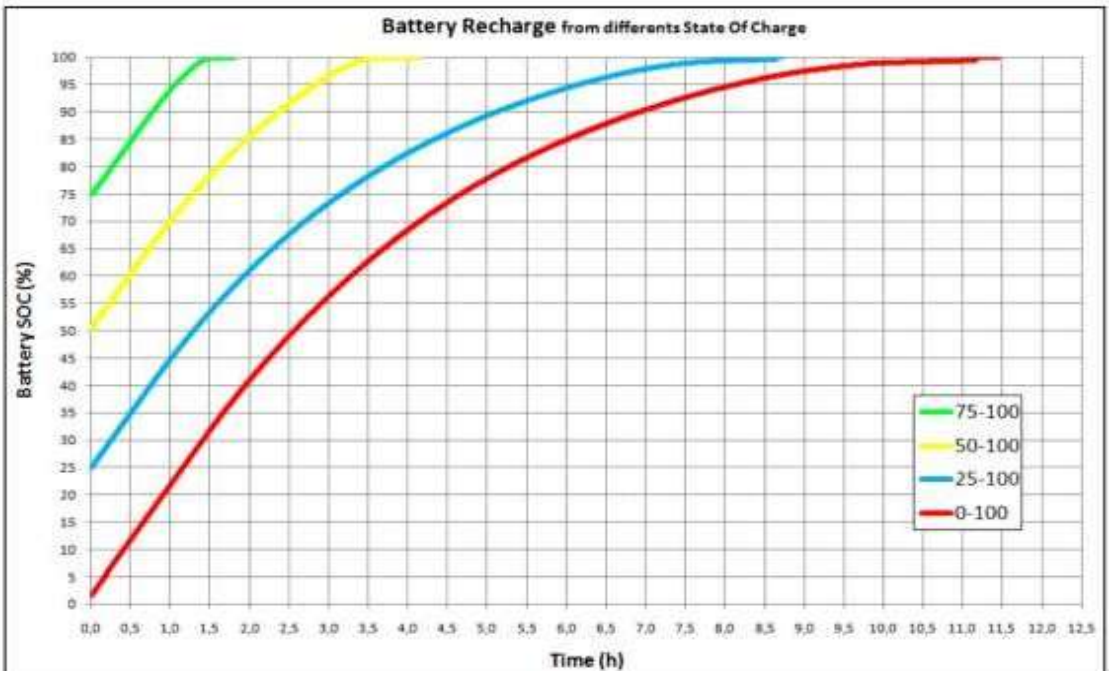
Number of battery modules	**64**

Battery Module Type	**250UP40**

Cell in series	100
Nominal Capacity [Ah]	40
Available Energy [Wh]	9600

DOD [%]	100%
Vmin [V]	210,00
Nominal Energy Output [Wh]	524720
Battery Energy Required [Wh]	579737

NOTE:



The graph below shows the recharge time for various initial state of charges. These results can be obtained when the recharge power required by the battery is fully available from the rectifier